Lecture 13: Related Rates I

Garrett Credi

10/22/2026

Exam 2 Soon!

Exam 2 is coming up! It is two Tuesday's from now, 11/3.
It's covering linearizations + our derivative rules + related rates + L'Hôpital's Rule which we'll cover next week.

Quiz Time

Let's get into neat rows, spread out, and slay this quiz!

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$$2y' = -10 \quad y' = -5 \tag{6}$$

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(Notice from before, we found the slope of the line tangent to the curve at $(1, 0)$ was -5 and $-15 = -5 \cdot 3$, so we're exactly enforcing that the velocity is parallel to the tangent line).

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3. Solving the resulting equation(s) for the rate of change in question.

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For a classic related rates problem: Suppose you have a cylindrical tank of water whose height is 10 meters and whose radius is 2 meters, with a hole at the bottom from which water drains out. At the moment the height of the water is halfway to the bottom, a hole at the bottom of the tank opens and you measure the speed at which the water escapes to be about $\frac{1}{2}\frac{m^3}{sec}$. At what rate will the height of the water decrease?

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$$\frac{dV}{dt} = \pi \cdot (\frac{dr}{dt} \cdot (\frac{d}{dr}r^2) \cdot h + r^2 \cdot \frac{dh}{dt}\frac{d}{dh}h) \tag{19}$$

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Related Rates Example I cont.

We then want to differentiate the relation (volume in terms of radius and height) with respect to time to relate the rates of change:

$$V = \pi r^2 h \tag{15}$$

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So the height of the water is decreasing at a rate of $\frac{1}{8\pi}$ meters per second.

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Giving us the *relation*:

$$V = \frac{1}{3}\pi r^2 h \tag{22}$$

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$$V = \frac{4\pi h^3}{300} = \frac{\pi h^3}{75} \tag{24}$$

Related Rates Example II cont.

We would then differentiate this relation w.r.t. t to get:

$$\frac{d}{dt}V = \frac{d}{dt}\frac{\pi h^3}{75} \tag{25}$$

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Related Rates Example II cont.

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$$\frac{dh}{dt} = \frac{25}{\pi h^2}\frac{dV}{dt} = \frac{25}{\pi \cdot 5^2} \cdot \frac{-1}{2} = \frac{-1}{2\pi}\frac{m}{s} \tag{27}$$

So now the height is decreasing at a rate of $\frac{1}{2\pi}$ meters per second.

Related Rates Example III

For a non-tank draining problem, let's instead suppose there is a 10 meter long ladder propped up at an angle on a wall that makes a right angle with the ground. It's base is currently 6 meters away from the wall, and that base is sliding away from the wall at a rate of two meters per second. At that time, how high up along the wall is the ladder, and what is the rate of change of that height?

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- RoC information: $\dfrac{dx}{dt} = 2\dfrac{m}{s}$.
- Information in question: value of $y(t)$ and $\dfrac{dy}{dt}$.

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We differentiate the relation w.r.t time:

$$x^2 + y^2 = 10^2 \tag{28}$$

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Related Rates Example III cont. 2

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and distances must be positive so $y(t) = 8$ when $x(t) = 6$.

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So:

$$\frac{dy}{dt} = -\frac{12}{y} = -\frac{12}{8} = -\frac{3}{2}\frac{m}{sec}. \tag{40}$$

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So:

$$\frac{dy}{dt} = -\frac{12}{y} = -\frac{12}{8} = -\frac{3}{2}\frac{m}{sec}. \tag{40}$$

The height of the ladder against the wall is therefore decreasing at a rate of $\frac{3}{2}$ meters per second.

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Generally however, all of these problems will rely on some kind of geometric input, e.g.:

- Area of square, rectangle, circle, ellipse, triangle
- Perimeter of the above (except ellipse)
- Angles (e.g. SOHCAHTOA trig)
- Volumes of cubes, rectangular prisms, pyramids, cones, spheres
- Surface areas of the cubes, rectangular prisms, spheres
- Pythagorean theorem
- Similar triangles

and thus having some way to commit these formulae to memory will be useful.

Worksheet Time

Let's get into groups and work on the worksheet!

End of Class

That’s all folks!